

Introduction: panel formation before blind
evaluation

Blind evaluation begins before the estimator I

Blind evaluation begins before the estimator II

Evaluating decision-makers without an answer key is a recurring problem: in crowdsourcing, in ensembles of classifiers, in deliberative bodies, and in any setting where the truth is expensive, contested, or unavailable at evaluation time — the problem Dawid and Skene first formalized for observer error-rate estimation without ground truth Dawid and Skene (1979). Later learning-from-crowds work made the same inferential problem operational for noisy human labels and missing gold standards Raykar et al. (2010), and budget-optimal crowdsourcing work shows why worker reliability, redundancy, and task assignment cannot be separated when answers are inferred from noisy repeated judgments rather than gold labels Karger et al. (2014). A parallel line estimates classifier accuracy *without any labels* by exploiting the structure of agreements among multiple predictors — logic- and constraint-based [platanios2014] and spectral [parisi2014] — and, like the exact NTQR solver, these methods lean on an **error-independence** (or low-rank residual-dependence) assumption whose violation is exactly the failure mode we make tunable here. The ntqr package (v0.8) frames the problem as

Application review as the empirical stress test I

Application review as the empirical stress test II

Application review is the manuscript's concrete empirical stress test because it combines expert judgment, scarce labels, panel formation, and plausible nuisance bias in one setting. Studies of academic and grant peer review emphasize that expert panels are not transparent measurement devices: judgment is field-situated Lamont (2009), replication of review decisions can be fragile Peters and Ceci (1982), reviewer agreement can be low even on the same proposals Cole et al. (1981); Pier et al. (2018), and statistical analysis of NIH review scores shows that panel-scoring uncertainty can materially change which proposals would be funded Johnson (2008). Productivity follow-up work adds a separate caution: NIH peer-review percentile scores can be weak predictors of grant productivity Fang et al. (2016), and peer-review bias is a documented design concern rather than an exotic failure mode Lee et al. (2013); Tomkins et al. (2017); Helmer et al. (2017). The postdoctoral fellowship version is also historically apt: Wennerås and Wold's analysis of postdoctoral fellowship review found nepotism and sexism in peer review Wennerås and Wold (1997), while broader funding studies report racial disparities in award

Sortition from civic lotteries to evaluator sampling I

Sortition — selection by lottery, often with quotas that make the drawn body mirror the population — is the canonical *fair*, non-comparative panel-formation rule Stone (2011). Modern implementations use auditable maximin algorithms Flanigan et al. (2021) that maximize the minimum selection probability subject to representativeness constraints. Sortition is attractive precisely because it does **not** select on competence; it selects on representativeness. That makes it a sharp test case for NTQR: a representative panel is heterogeneous and, by design, *not* curated for accuracy. Does representativeness help or hurt an estimator that depends on the statistical independence of judges' errors?

Sortition from civic lotteries to evaluator sampling II

That procedural tension has a long pre-1800 lineage. Aristotle's political theory treats lot and election as constitutional signals — lot as democratic, election as oligarchic — while the Athenian institutional account describes juries and offices allocated by lot Aristotle (Politics); Aristotle (Athenian Constitution). Medieval and early-modern writers kept the same problem visible under different vocabularies: Aquinas (Summa Theologiae II-II q.95 a.8) distinguished practical uses of lots from divinatory misuse, and Contarini (1599) described Venetian mixed selection machinery as an anti-factional civic design. Enlightenment writers then made the lot-versus-choice contrast explicit again: Montesquieu (1748) and Rousseau (1762) both distinguish election by lot from election by choice. We cite these sources as procedural history, not as direct empirical precedent: this manuscript does not claim that Athenian juries, Venetian offices, or scholastic accounts of chance anticipate NTQR. They do show that the upstream choice between lot, choice, status, and asserted competence is an old institutional design problem.

Falsifiable claims and negative controls I

We state the study as five falsifiable hypotheses (H1–H5) and let the regenerated artifacts adjudicate each. The synthetic deterministic track tests H1–H4 against a known oracle; the live single-model companion tests H5. Methods (Table @tbl:falsification) states the load-bearing assumption and the negative-control check behind each, and the Discussion returns an explicit verdict hypothesis by hypothesis.

1. **H1 — Formation strategy is the dominant lever.**

Different panel-formation rules yield materially different oracle-referenced EIE recovery error, even at a fixed population and panel size. *Tested by* the weighted-mean strategy ranking with a bootstrap separation gate and a power budget ([@fig:ranking]).

Falsifiable claims and negative controls II

2. H2 — **Concentrating correlated bias degrades recovery.**

Single-bloc selection, which seats judges whose errors are correlated, recovers with higher EIE error than a representative draw, because the error-independence assumption NTQR rests on is most stressed by a correlated panel. *Tested by* the representative-minus-single-bloc contrast across the full regime grid against analytical sign predictions ([@fig:repideo]), and resolved by the composition-coupled error confound that fans the strategies apart as within-bloc coupling rises ([@fig:blocphase]).

Falsifiable claims and negative controls III

3. **H3 — Size is a sampling knob, not guaranteed improvement.** Forming a larger ensemble gives more trios to average over, but whether that helps or hurts recovery — and whether any effect comes from the larger ensemble violating the solver's error-independence assumption more — is an empirical question; the power question is how many observations at the analyzed grain would be needed to resolve a contrast of the observed magnitude. *Tested by* the per-strategy size sweep, a paired regime-controlled size contrast, a per-trio conditioning diagnostic, and the power/MDE budgets ([@fig:powercurve], [@fig:powervn], [@fig:triocond]).

Falsifiable claims and negative controls IV

4. **H4 — Error-correlation is measurable, and recovery degrades with it.** A controlled correlation injection produces a realized error-correlation that NTQR itself reports, and oracle-referenced recovery error should rise as that realized correlation grows. *Tested by* the tolerance sweep, reporting the realized-correlation trend and an OLS recovery-vs-correlation slope with a bootstrap interval ([@fig:tolerance]); the recovery slope, unresolved by the global-injection sweep, is resolved positive once the correlation is composition-coupled and marginal-preserving ([@fig:blocphase]).

Falsifiable claims and negative controls V

5. **H5 — The synthetic ranking transfers to a live single-model panel.** The strategy ordering measured against the synthetic oracle reproduces when one local gemma3:4b model is prompted as different reviewers. *Tested by a matched-grain cross-track ranking comparison and cell-level directional alignment ([@fig:trackinversion], [@fig:postdocalign]).*

These hypotheses are written to be refutable, and the data refute several: H3 and H5 are rejected; H2 and H4 are unresolved on the baseline grid and resolve only once correlation is coupled to panel composition; and H1's ranking collapses to competence-first versus a bunched remainder. The instrument, its resolved cells, its explicit design-limited cells, and the axis-conditional negative control that keeps the sortition result honest — not a manufactured sortition win — are the contribution.